Topic Classification for Kinyarwanda News:

Sparse, Recurrent and Transformer Models on a Deduplicated KINNEWS Benchmark

Elvin Cyubahiro

[[TODO: course name / institution]]

Summative Project: Text Classification for an African Language · October 2026

Project links

Resource	Link
GitHub repository	[[TODO: GitHub repository URL]]
Demo video	[[TODO: demo video URL]]
Live system	[[TODO: deployed web application URL]]

Abstract

Kinyarwanda is spoken by an estimated 30 million people (Niyongabo et al., 2020) but remains poorly served by NLP resources (Joshi et al., 2020). This project builds and deploys a 14-class topic classifier for Kinyarwanda news using the KINNEWS corpus (Niyongabo et al., 2020). An audit of the corpus found that 55.2% of the training file consists of duplicated articles and that the official train and test splits have markedly different class distributions; the training set was therefore deduplicated (17,014 to 7,626 articles) before a stratified validation split was drawn, and the official test set was left untouched. Nine configurations were compared under identical splits: a majority-class floor, TF-IDF word n-grams with logistic regression (the baseline), TF-IDF character n-grams, a BiLSTM with random and with fastText initialisation, and fine-tuned XLM-R and AfroXLMR, plus two ablations. Using macro-F1 as the primary metric, the character n-gram model performed best (0.587 test macro-F1, 95% CI 0.571–0.603), beating the word baseline by 7.6 points with half the parameters. fastText initialisation improved the BiLSTM by 6.7 points, and African-centric adaptation (AfroXLMR) improved on XLM-R by 6.3 points, all significant under McNemar's test. Neither neural family surpassed the character n-gram model in this setting. Error analysis attributes most remaining errors to overlapping topics, very small training classes, and label noise in the corpus. The main conclusion is that, for a small and morphologically rich language corpus, representing sub-word structure matters more than model size.

1. Introduction

Kinyarwanda is an official language of Rwanda and, according to figures cited by Niyongabo et al. (2020), is spoken by approximately 30 million people across Rwanda, Uganda, the Democratic Republic of Congo and Tanzania. Despite this, it sits in the long tail of languages with little labelled data or tooling (Joshi et al., 2020). Rwandan outlets publish news in Kinyarwanda every day, and assigning each article to a topic is a routine task for newsrooms, archives and aggregators. Automating it saves editorial time, makes archives searchable by subject, and lets readers and downstream systems filter content.

The task is also a good test of how NLP methods behave on an African language. Kinyarwanda is a **morphologically rich** Bantu language (Nzeyimana & Niyongabo Rubungo, 2022) with a **system of noun classes** marked by prefixes (Kimenyi, 1980): a single stem appears in many written forms depending on its prefixes and affixes (for example *umukino*, *imikino*, *abakinnyi* around the stem for “play/game”). A word-level model sees each form as a separate feature, which spreads the evidence for one concept thinly. This property motivates the research questions below.

Intended users are newsroom and archive staff tagging incoming articles, content aggregators routing news by topic, and researchers or developers who need a documented Kinyarwanda classifier.

Objective. Build, evaluate and deploy a classifier that assigns a Kinyarwanda news article to one of 14 topics, and use controlled experiments to answer four research questions:

RQ	Question
RQ1	How far can a sparse linear model go, and do character n-grams beat word n-grams?
RQ2	Do subword (fastText) embeddings trained on in-domain text help a BiLSTM over random initialisation?
RQ3	Does African-centric pretraining (AfroXLMR) beat general multilingual pretraining (XLM-R)?
RQ4	What does the best model still get wrong, and why?

2. Related Work

2.1 NLP for low-resource African languages

Joshi et al. (2020) show that NLP resources are distributed extremely unevenly: a small number of languages hold almost all labelled data and benchmarks, while languages such as Kinyarwanda, despite millions of speakers, have very little; Niyongabo et al. (2020) note that this taxonomy places both Kinyarwanda and Kirundi in the “Scraping-By” category. In this setting a corpus of a few thousand labelled articles is a substantial

resource, and methods that generalise from few examples matter more than on English benchmarks. Ogueji et al. (2021) challenge the assumption that low-resource languages need very large multilingual models, showing that a model pretrained on a small, focused African-language corpus can compete with much larger general models.

2.2 KINNEWS and prior Kinyarwanda classification

Niyongabo et al. (2020) introduced KINNEWS and KIRNEWS, paired news classification corpora for Kinyarwanda and the closely related Kirundi, with the aim of enabling cross-lingual transfer between two under-resourced Bantu languages. They reported monolingual and cross-lingual baselines: multinomial Naive Bayes, logistic regression and a linear SVM over TF-IDF features, a character-level CNN, and a CNN and a BiGRU over Word2Vec embeddings. This project uses the Kinyarwanda portion. Its results are not directly comparable with the original baselines, because the training set here was deduplicated (Section 3) and the primary metric is macro-F1 rather than accuracy.

2.3 Sparse and recurrent text classification

Logistic regression over TF-IDF features is a fast, interpretable and strong baseline for topic classification (Pedregosa et al., 2011). Character-level modelling has already been applied to Kinyarwanda: Niyongabo et al. (2020) include a character-level CNN among their baselines. Bojanowski et al. (2017) carried the same idea into dense representations: fastText extends Word2Vec (Mikolov et al., 2013) by composing each word vector from its character n-grams, so related forms share structure and unseen words still receive a vector. Recurrent models based on the LSTM (Hochreiter & Schmidhuber, 1997), run in both directions (Schuster & Paliwal, 1997), were the standard neural text classifiers before Transformers. Most directly relevant to this project, Nzeyimana and Niyongabo Rubungo (2022) argue that the unsupervised subword tokenization used by multilingual models, such as byte-pair encoding, is sub-optimal for morphologically rich languages, and propose KinyaBERT, a Kinyarwanda language model that uses a morphological analyser to represent words explicitly as compositions of morphemes.

2.4 Multilingual and African-centric Transformers

The Transformer (Vaswani et al., 2017) and the pretrain-then-fine-tune paradigm (Devlin et al., 2019) now dominate text classification. XLM-R (Conneau et al., 2020) is a multilingual encoder pretrained with masked language modelling on the CC-100 web corpus. Checked directly for this project, Kinyarwanda (“rw”) is absent from CC-100’s 116 language files and from the 94 languages listed on the xlm-roberta-base model card, although related Bantu languages (Swahili, Zulu, Xhosa) are included. Alabi et al. (2022) proposed multilingual adaptive fine-tuning (MAFT): continuing masked-language-model training of an existing model on a set of African languages. The resulting

AfroXLMR-base covers 17 African languages, and its model card lists Kinyarwanda. Because AfroXLMR reuses XLM-R's tokenizer unchanged, comparing the two isolates the effect of adapted weights from the effect of tokenization.

2.5 Gap addressed

This project introduces no new dataset or architecture. Its contribution is a controlled comparison on KINNEWS of sparse word versus sparse character features, a recurrent model with and without subword embeddings, and a general versus an African-adapted Transformer, all on identical deduplicated splits, with confidence intervals, paired significance tests, a detailed error analysis and a deployed tool. It also documents two properties of the benchmark (a 55% duplicate rate and a strong train/test class shift) that change how any result on it should be read.

3. Dataset and Data Preparation

3.1 Source and licence

KINNEWS (Niyongabo et al., 2020) contains Kinyarwanda news articles collected from fifteen news websites and five newspapers in Rwanda, released under the MIT licence. Each article has a headline, its content and the source URL. The raw version was used (title and full article text) rather than the pre-cleaned version, so that the Transformer models receive natural text and so that preprocessing is documented in this project. The loader in the notebook tries the Hugging Face dataset first; that route no longer works with current versions of the datasets library, so the corpus is read directly from the authors' release archive, whose SHA-256 hash is recorded to pin the exact version. All published statistics were confirmed programmatically: 17,014 training and 4,254 test articles, 14 classes, and the six documented fields.

How the categories were defined. According to Niyongabo et al. (2020), the publishers had originally filed the articles under 48 different categories. Because many were closely related, the dataset's annotators merged them into 14 topics: politics, sport, economy, health, entertainment, history, technology, tourism, culture, fashion, religion, environment, education and relationship. Two annotators, both linguistics graduates and native speakers, then reviewed each article and assigned it one of these topics based on its title and content; articles they were unsure about were given a neutral label and removed. Each topic has a numeric label (1–14), an English label and a Kinyarwanda label. This project uses these labels as published; Section 3.2 shows that, despite this process, the released training file still contains conflicting labels for identical articles.

3.2 Quality audit

A systematic audit before any modelling found four issues that shaped the rest of the project:

1. **Heavy duplication.** Only 7,629 of 17,014 training rows carry distinct text; 55.2% are copies, and one article appears 80 times. If duplicates are not removed before the validation split is drawn, copies of the same article land in both training and validation, and validation measures memorisation.
2. **Label noise.** 156 groups of identical articles carry more than one label (1,446 training rows). No model can be right about both copies, which places a ceiling on achievable accuracy.
3. **Train/test distribution shift.** The official splits do not share a class distribution. Environment is about 19 times more frequent in the test set than in the raw training set, and fashion about 11 times; economy is about five times less frequent. The total variation distance between the two distributions is 0.383 (0.320 after deduplication). Figure 1 shows the ratios after deduplication.
4. **Train/test leakage.** Three training articles were verbatim copies of test articles and were removed; 61 test articles (1.4%) have a near-duplicate (TF-IDF cosine > 0.9) in training. These were reported, not removed, because the test set is never modified.

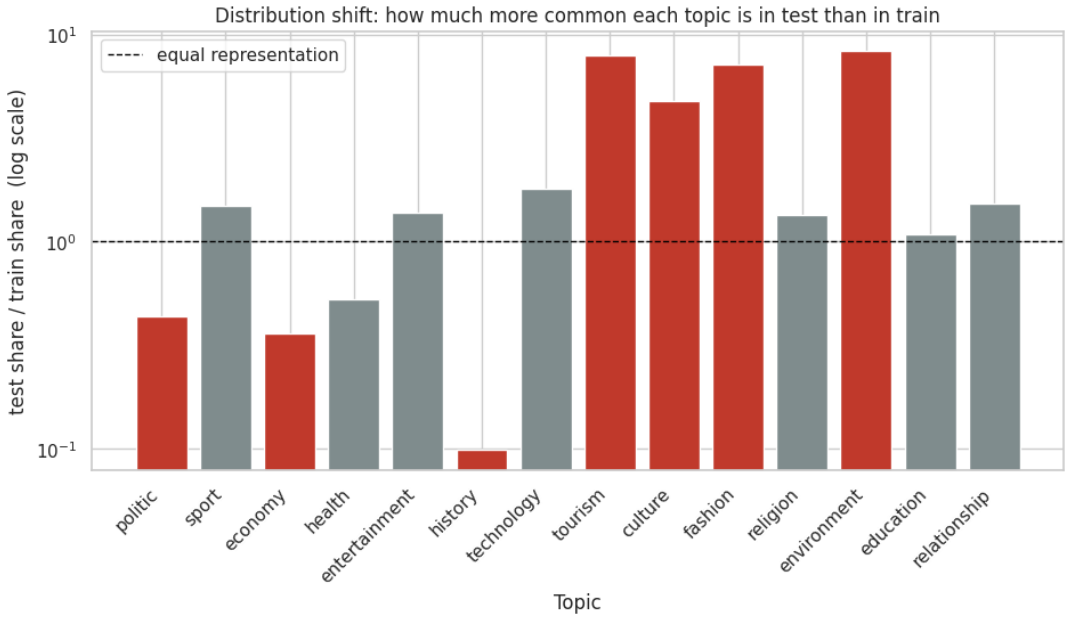


Figure 1. How much more common each topic is in the test set than in the deduplicated training set (log scale). Red bars differ from equal representation by more than a factor of two.

3.3 Characteristics

Articles are long: the median training article has 292 words, and 716 XLM-R subword tokens (Figure 2). 94.9% of articles exceed 256 subword tokens and 74.5% exceed 512, so any Transformer with a 512-token limit never sees the end of three-quarters of the

corpus. Vocabulary is highly sparse: the corpus contains 231,516 distinct word types in 2.38 million tokens, and **61.6% of word types occur exactly once**. Both the XLM-R and AfroXLMR tokenizers split Kinyarwanda into 2.47 subword tokens per word on average, and produce identical output.

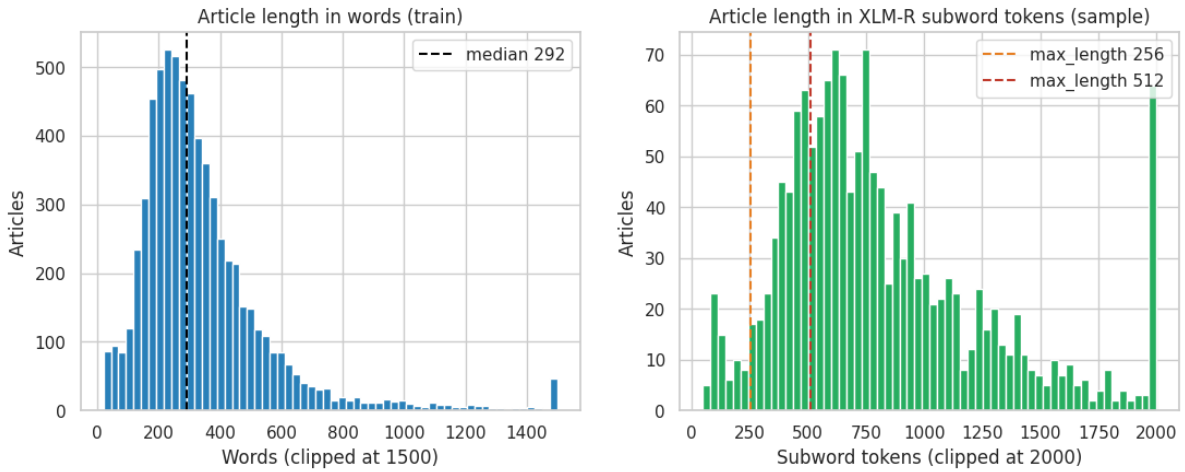


Figure 2. Article length in words (left) and in XLM-R subword tokens (right). Dashed lines mark the 256 and 512 token limits used by the Transformer models.

3.4 Deduplication and splits

Exact duplicates (after Unicode normalisation) were removed from the training file, keeping the first occurrence, which reduced it from 17,014 to 7,626 articles. This also reduced the class imbalance ratio (largest to smallest class) from 117.5 to 53.2, because duplicates were concentrated in the largest classes. A stratified 10% validation split was then drawn with a fixed seed (42). The official test split was used as published. Split indices are saved to data/splits.json, and the notebook asserts that no article text is shared across the three splits.

Table 1. Data splits after deduplication.

Split	Articles	Use
Train	6,863	Model fitting (deduplicated, stratified 90%)
Validation	763	All hyperparameter choices, early stopping, model selection, app threshold
Test	4,254	Official split, untouched; evaluated once per final model

The smallest classes are very small after deduplication: tourism has 31 training articles, environment and fashion 34 each, and culture 69, while the test set contains 150–200 articles of each. Class weighting is therefore essential rather than optional.

3.5 Preprocessing

Two preprocessing pipelines were used, matched to what each model family expects:

- **Classical and BiLSTM models:** Unicode NFC normalisation, lower-casing, replacement of URLs, e-mail addresses and numbers with placeholder tokens, removal of punctuation except apostrophes (which carry meaning in Kinyarwanda contractions such as *n'abandi*), and whitespace normalisation.
- **Transformers:** Unicode NFC normalisation and whitespace normalisation only, so the input stays close to the natural text the models were pretrained on.

In both cases the model input is the title followed by the article body.

4. Methodology

All models were trained on the same deduplicated training split, tuned only on the validation split, and evaluated once on the test split. Class imbalance was handled with balanced class weights for the linear models and a class-weighted cross-entropy loss for the BiLSTM. The complete implementation is a single notebook that can be run end to end on Kaggle or Google Colab with one GPU.

4.1 Baselines

- **B0 – majority class.** Always predicts the most frequent training class (politics). It sets the floor and shows why accuracy is misleading on this data.
- **B1 – TF-IDF word n-grams + logistic regression (the baseline).** Word unigrams and bigrams (minimum document frequency 2, up to 200,000 features, sublinear term frequency), fed to a multinomial logistic regression with balanced class weights. The regularisation strength C was selected from {0.1, 1, 10} on validation (C = 1). This is the standard strong baseline for topic classification.

4.2 Experiments

- **E1 – TF-IDF character n-grams + logistic regression (RQ1).** Identical to B1 except for the features: character 2–5-grams within word boundaries (char_wb, minimum document frequency 3). Only the feature type changes, so any difference is attributable to it.
- **E2a / E2b – BiLSTM (RQ2).** A bidirectional LSTM written from scratch in PyTorch: 300-dimensional word embeddings, one layer of 128 hidden units per direction, masked mean pooling over the non-padding positions, dropout 0.3 and a linear layer to 14 classes. Inputs are truncated to 400 words; the vocabulary (72,309 types, minimum frequency 2) is built on the training split only. Training uses Adam (Kingma & Ba, 2015; learning rate 0.001), batch size 64, gradient clipping at 1.0, class-weighted cross-entropy, and early stopping on validation macro-F1 (patience 3, up to 15 epochs). E2a initialises the embeddings randomly. E2b initialises them from fastText skip-gram vectors (300 dimensions, character n-grams of length 3–6, window 5, 10 epochs) trained on the training split only; all

72,307 vocabulary words receive a pretrained vector. Everything else is identical.

- **E3 / E4 – fine-tuned Transformers (RQ3).** xlm-roberta-base (E3) and Davlan/afro-xlmr-base (E4), each with the standard sequence-classification head on the first-token representation, fully fine-tuned (278 million parameters) with AdamW (Loshchilov & Hutter, 2019), learning rate 2e-5, linear schedule with 10% warm-up, weight decay 0.01, batch size 16, 3 epochs, mixed precision, and a maximum length of 512 tokens. The checkpoint with the best validation macro-F1 is kept. The two runs differ only in the pretrained weights, since the tokenizers are identical.

4.3 Ablations

Both ablations modify the stronger Transformer (E4) so that a single factor changes:

- **A1 – title only.** Trains and evaluates on the headline alone, measuring how much topic signal the headline carries.
- **A2 – maximum length 256.** Halves the input length, measuring what the second 256 tokens contribute given the truncation documented in Section 3.3.

4.4 Evaluation protocol

Macro-F1 is the primary metric: it averages F1 over the 14 classes, so every topic counts equally regardless of size. This matters here because accuracy rewards predicting the large classes; B0 reaches 10.4% accuracy on test but only 1.3% macro-F1. Accuracy, weighted F1, and per-class precision, recall and F1 are also reported. Uncertainty on the test score is estimated with a 95% bootstrap confidence interval over 1,000 resamples of the test set (Efron & Tibshirani, 1993). Pairs of models are compared with McNemar's test on their paired correct/incorrect predictions (McNemar, 1947), using the continuity-corrected chi-squared form (exact binomial when fewer than 25 predictions disagree). The final model was selected on validation macro-F1, never on test.

4.5 Implementation and resources

The project is implemented in Python with scikit-learn (Pedregosa et al., 2011), PyTorch (Paszke et al., 2019), gensim (Řehůřek & Sojka, 2010) and Hugging Face Transformers (Wolf et al., 2020); the app uses Gradio (Abid et al., 2019). Library versions are pinned in requirements.txt. All reported results come from one complete run on a Kaggle notebook with a single NVIDIA T4 GPU (about 50 minutes end to end). Every experiment writes a JSON result file and a per-article prediction file, and every table and finding in the notebook is generated from those files rather than typed by hand.

Existing resources versus own work. The dataset (KINNEWS), the pretrained checkpoints (xlm-roberta-base, Davlan/afro-xlmr-base) and the libraries above are

existing resources. The data audit and deduplication, preprocessing, the BiLSTM architecture and training loop, the fastText training, the fine-tuning and evaluation code, the significance testing, the error analysis and the web application were implemented for this project. **[[TODO: state your use of AI assistance here if your course requires it]]**

5. Experiments and Results

Table 2 lists every configuration, sorted by validation macro-F1, which is the criterion used to select the final model.

Table 2. All experiments (Kaggle run, single seed). The selected model is highlighted. CI: 95% bootstrap confidence interval.

ID	Model	RQ	Val macro-F1	Test macro-F1	95% CI (test)	Test acc.	Params
E1	TF-IDF char 2–5-grams + LR	RQ1	0.700	0.587	0.571–0.603	0.726	1.36 M
E2b	BiLSTM + fastText	RQ2	0.664	0.509	0.492–0.525	0.675	22.1 M
B1	TF-IDF word 1–2-grams + LR	baseline	0.639	0.512	0.494–0.528	0.693	2.80 M
E2a	BiLSTM, random init.	RQ2	0.573	0.442	0.424–0.458	0.591	22.1 M
E4	AfroXLMR-base, fine-tuned	RQ3	0.523	0.457	0.445–0.468	0.714	278 M
A2	E4, max length 256	ablation	0.518	0.445	0.434–0.456	0.701	278 M
A1	E4, title only	ablation	0.436	0.411	0.399–0.422	0.663	278 M
E3	XLM-R-base, fine-tuned	RQ3	0.398	0.394	0.384–0.403	0.672	278 M
B0	Majority class	floor	0.027	0.013	0.012–0.015	0.104	–

5.1 RQ1: character n-grams versus word n-grams

E1 reaches 0.587 test macro-F1 against B1's 0.512, a gain of **7.6 points** with non-overlapping confidence intervals and **less than half the parameters** (1.36 M versus 2.80 M). McNemar's test confirms the difference is systematic: E1 is right on 278 articles that B1 gets wrong, and B1 on only 135 that E1 gets wrong ($p = 2.8 \times 10^{-12}$). The gain is concentrated in the smallest training classes: environment (34 training articles) improves by 0.295 F1, tourism by 0.234, fashion by 0.201 and culture by 0.172, while the large classes barely move. This fits the morphological argument: character n-grams let the model pool evidence across the many inflected forms of a stem, which matters most when a class has few examples. The highest-weighted E1 features confirm this: they are stems and affixes such as "kipe" (team), "uzima" (health) and "dukik" (from *ibidukikije*, environment) rather than whole words.

5.2 RQ2: fastText initialisation for the BiLSTM

Initialising the BiLSTM with fastText vectors raised test macro-F1 from 0.442 (E2a) to 0.509 (E2b), +**6.7 points** with non-overlapping confidence intervals (McNemar $p = 3.9 \times 10^{-33}$). The largest gains were in culture (+0.19 F1, 69 training articles), politics, technology and religion. The fastText vectors behave as intended: the nearest neighbours of *ibidukikije* (environment) are its inflected forms (*kubidukikije*, *y'ibidukikije*, *n'ibidukikije*), and an invented out-of-vocabulary form, *ibidukikijebyacu*, still receives a vector whose neighbours are the same family. However, even with this help the BiLSTM does not match the character n-gram model (0.509 versus 0.587), and both BiLSTMs overfit quickly: training loss falls towards zero while validation loss starts rising after five to nine epochs. With 22 million parameters (21.7 million in the embedding layer) and fewer than 7,000 training articles, the recurrent model has far more capacity than the data can support.

5.3 RQ3: African-centric versus general multilingual pretraining

AfroXLMR (E4) outperformed XLM-R (E3) by **6.3 points** (0.457 versus 0.394), with non-overlapping confidence intervals and McNemar $p = 3.2 \times 10^{-17}$. Because the two models share an identical tokenizer and identical fine-tuning settings, the difference is attributable to the adapted weights, i.e. to AfroXLMR having seen Kinyarwanda during adaptation (Alabi et al., 2022). The gap is largest on technology (+0.21 F1) and history, where XLM-R never predicted the class correctly (F1 = 0.00). AfroXLMR also learned faster, reaching 0.499 validation macro-F1 after one epoch against XLM-R's 0.260.

Both Transformers nevertheless scored well below the character n-gram model, which is the most important negative result of the project. Four factors plausibly explain it:

- **No class weighting.** The linear models and BiLSTMs were trained with class weights, but the Transformers used the standard unweighted loss. With classes of 31–34 training articles, the Transformers under-predict the rare topics that macro-F1 weights equally. This is a confound in the comparison, and the first thing to fix in future work.
- **Truncation.** Three-quarters of articles are longer than 512 tokens (Section 3.3), so the Transformers see only the beginning of most documents, while the TF-IDF models use the whole text.
- **Limited training.** Three epochs on 6,863 articles with a single learning rate; validation macro-F1 was still rising at the end of training for both models, suggesting they were under-trained.
- **Tokenizer fit.** The shared tokenizer was built without Kinyarwanda and splits each word into about 2.5 pieces, which adaptation of the weights cannot change. Nzeyimana and Niyongabo Rubungo (2022) identify exactly this kind of subword tokenization as sub-optimal for Kinyarwanda.

Accuracy tells a slightly different story: E4's test accuracy (0.714) is close to E1's (0.726), so AfroXLMR handles the frequent classes well and loses mainly on the rare ones. This is consistent with the class-weighting explanation.

5.4 Ablations

Title only (A1). Using only the headline reduced E4's test macro-F1 from 0.457 to 0.411 (−4.6 points, significant). Headlines therefore carry most, but not all, of the topic signal; health and technology lost the most. This is useful for deployment, since users often paste only a headline. **Max length 256 (A2).** Halving the input length cost only 1.1 points (0.457 to 0.445), with overlapping confidence intervals, although McNemar's test still detects a systematic difference ($p = 8.2 \times 10^{-6}$). The second 256 tokens add little, which suggests that simply extending the window further would help less than better training.

5.5 Validation versus test

Every model scores lower on test than on validation (E1: 0.700 to 0.587). This is expected here, because validation follows the training class distribution while the official test set does not (Figure 1): the test set contains many more articles from the rarest training classes, which are exactly the classes every model finds hardest.

5.6 Inference cost

Measured on the same machine with batched inference, E1 classifies an article in about 0.03 ms, the BiLSTMs in about 0.36 ms, and the full Transformers in about 8.7–8.9 ms. E1 is therefore both the most accurate model on macro-F1 and roughly 270 times cheaper to run than the Transformers, which supports deploying it.

6. Error Analysis and Discussion

The error analysis focuses on the selected model, E1. Table 3 shows its per-class performance on the test set.

Table 3. E1 per-class test performance, sorted by F1 (macro-F1 0.587, weighted F1 0.730, accuracy 0.726).

Class	Train n	Test n	Precision	Recall	F1
sport	878	812	0.955	0.925	0.940
entertainment	994	848	0.828	0.824	0.826
relationship	380	360	0.722	0.947	0.820
religion	401	335	0.835	0.740	0.785
health	978	320	0.619	0.741	0.674
technology	136	152	0.539	0.822	0.651
tourism	31	152	0.827	0.533	0.648

Class	Train n	Test n	Precision	Recall	F1
politics	1,628	442	0.699	0.584	0.636
fashion	34	151	0.852	0.497	0.628
economy	1,000	222	0.488	0.662	0.562
environment	34	175	0.927	0.291	0.443
culture	69	203	0.656	0.300	0.412
education	104	70	0.179	0.071	0.102
history	196	12	0.048	0.833	0.091

6.1 Where the model works

Classes with distinctive vocabulary are handled well even when small. Sport reaches 0.94 F1, driven by unambiguous stems such as *ikipe* (team), *umukino* (match) and club names. Relationship, religion and entertainment are also above 0.78. Technology (136 training articles) and tourism (31) reach about 0.65, which shows that character n-grams can learn a topic from few examples when its vocabulary is specific (for tourism: *pariki*, *Nyungwe*, *ingagi* “gorillas”, and the tourism board RDB).

6.2 Where the model fails

Figure 3 compares the baseline and final confusion matrices. The most frequent confusions (Table 4) fall into a small number of recurring types.

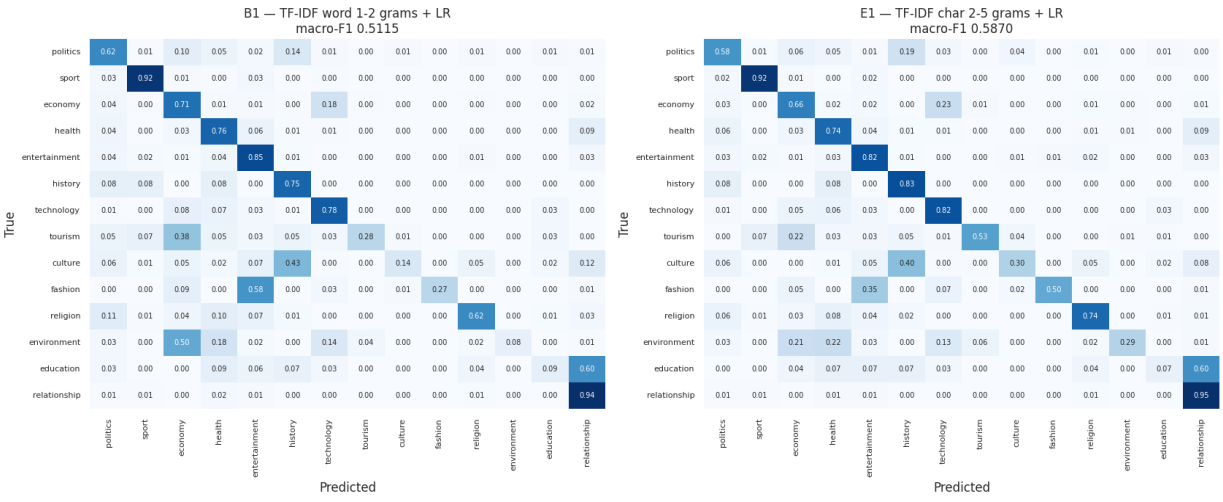


Figure 3. Row-normalised test confusion matrices for the baseline B1 (left) and the final model E1 (right). The diagonal is per-class recall.

Table 4. The most frequent confusions of E1 on the test set.

True class	Predicted	Articles	Share of true class
politics	history	85	19.2%
culture	history	82	40.4%
fashion	entertainment	53	35.1%
economy	technology	50	22.5%
education	relationship	42	60.0%
environment	health	39	22.3%
environment	economy	37	21.1%
tourism	economy	33	21.7%

1. **Genuine topic overlap.** Many errors are between topics that share vocabulary and often share subject matter. Politics articles about the 1994 Genocide against the Tutsi, or a biography of the President, are predicted as history; a culture article about Rwanda's last queen is predicted as history; fashion shows are predicted as entertainment ("Met Gala", "Victoria's Secret"). These are arguably ambiguous rather than wrong, and a multi-label scheme would describe them better.
2. **Over-prediction of history.** History has 196 training articles but only 12 test articles. With balanced class weights and a strong overlap with politics and culture, E1 predicts history 208 times, giving high recall (0.83) but precision of 0.05. This is a direct consequence of the train/test shift combined with class weighting.
3. **Very small classes.** Grouping test articles by the training size of their true class, accuracy is 79.1% for classes with 500 or more training articles, 78.5% for 100–499, but only **39.4% for classes with fewer than 100** (tourism, environment, fashion, culture). Environment illustrates the problem: its precision is high (0.93) but it is recalled only 29% of the time, because 34 training articles cannot cover the variety of environmental reporting, which is often framed in terms of health (pollution), economy (agriculture) or local government.
4. **Label noise and annotation choices.** The ten most confident mistakes are dominated by articles labelled entertainment that are clearly about football (e.g. a match draw between Rayon Sports and AS Kigali, predicted sport with 0.98 confidence) or about church events (predicted religion). Combined with the 156 duplicate groups carrying conflicting labels, this suggests a share of the "errors" are labelling inconsistencies in the source data rather than model failures. These cases are flagged as candidates for manual review in the notebook, not silently corrected.

5. **Education and scraping artefacts.** Education is the weakest class (F1 0.10): 60% of its test articles are predicted as relationship, often articles about children, parents and schooling. Its top word features also include “caption”, WordPress image markup left over from scraping, meaning the model partly learned how articles were scraped rather than what they say. A related artefact is visible in culture, whose features include “nâ” and “wâ”: curly apostrophes stored with the wrong character encoding in some source articles.
6. **Article length.** Performance is lowest for very short articles (macro-F1 0.52 below 100 words) and for very long ones (0.56 above 500 words), and highest between 250 and 500 words (0.60). Short texts carry little evidence, while long ones mix topics.

6.3 Qualitative examples

Table 5 shows real test headlines with the true label and E1’s prediction. In the first three E1 is right and the word-level baseline B1 is wrong; the rest are E1 errors.

Table 5. Example test headlines and predictions. English glosses are approximate.

Headline (Kinyarwanda)	Approximate English gloss	True	Predicted	Reading
Coronavirus: Ibitaramo bya Madonna na Miley Cyrus byahagaritswe	Coronavirus: Madonna and Miley Cyrus concerts cancelled	entertainment	E1: entertainment (B1: health)	Correct
Ghana: Umuhanzi ukunzwe mu ndirimbo ziririmbirwa Imana araregwa kubeshya ko avura SIDA	Ghana: a popular gospel singer is accused of falsely claiming to cure AIDS	entertainment	E1: entertainment (B1: health)	Correct
Ku rutonde rw’abantu bishimirwa cyane ku isi Bill Gates na Michelle Obama barayoboye…	Bill Gates and Michelle Obama top the list of the world’s most admired people…	entertainment	E1: entertainment (B1: sport)	Correct
Tombora yongeye guhuza Rayon Sports na AS Kigali	The draw has again paired Rayon Sports with AS Kigali	entertainment	sport (0.98)	Likely label noise
Shekinah Worship Team Masoro yamuritse album mu gitaramo cyamaze amasaha 12	Shekinah Worship Team Masoro launched an album at a 12-hour concert	entertainment	religion (0.93)	Topic overlap
Imyaka 26 irashize umwamikazi wa nyuma w’u Rwanda, Rosalie Gicanda yishwe	26 years since Rwanda’s last queen, Rosalie Gicanda, was killed	culture	history (0.72)	Topic overlap
Imyambarire idasanzwe yaranze ibirori bya Met Gala (Amafoto)	The unusual outfits of the Met Gala (photos)	fashion	entertainment (0.59)	Topic overlap

The successes share a pattern: the headline contains a word pointing to another topic ("Coronavirus", *SIDA* "AIDS", or a ranking that resembles sport coverage), which misleads the word-level model, while character n-grams from the rest of the text, such as *ibitaramo* (concerts) and *umuhanzi* (artist), carry the entertainment signal. The errors are the types described in Section 6.2: a football draw labelled entertainment in the corpus, and events that genuinely belong to two topics.

6.4 Out-of-domain behaviour

E1 assigns a topic to any input: English and French news sentences, Kirundi text, social-media Kinyarwanda and random characters all received a label. Its confidence on these inputs, however, was never higher than 0.24, well below typical in-domain confidence. A low-confidence threshold of **0.46** was therefore chosen on the validation set by maximising Youden's J statistic (Youden, 1950), treating correct and incorrect predictions as the two groups. Predictions above it are 89.5% accurate on validation and those below it 64.8% accurate; all the out-of-domain probes fall below it. The deployed app uses this threshold to warn users.

6.5 Relation to the literature

The results agree with the view that representing subword structure is central for morphologically rich, low-resource languages (Bojanowski et al., 2017; Nzeyimana & Niyongabo Rubungo, 2022): both of the project's subword mechanisms (character n-grams and fastText) gave the largest improvements, and they helped the smallest classes most. The RQ3 result supports Alabi et al. (2022): adapting a multilingual model to African languages clearly helps on a language the base model never saw. At the same time, the finding that a linear model beats both Transformers is in line with Ogueji et al. (2021): with limited data, model scale and general pretraining do not guarantee better performance, and choices such as class weighting and input length can matter more.

7. Deployment

The final model is exposed through a web application built with Gradio (Abid et al., 2019). A user pastes a Kinyarwanda headline or article and receives the three most likely topics with their probabilities. The app applies the same preprocessing as training, rejects empty input and inputs shorter than three words with an explanatory message, and shows a low-confidence warning when the top probability is below the validation-chosen threshold of 0.46. It includes example buttons drawn from real test articles, including a known failure (a football article labelled entertainment in the corpus), and an "About the model" tab with the metrics table and the main limitations.

The request flow is: input text → guard for empty or very short input → preprocessing → trained model → probability for each of the 14 topics → top three topics and confidence

warning. Because E1 is small (a vectoriser and a linear model of a few megabytes) and fast (well under a millisecond per article), it can be served on a free CPU instance without a GPU.

[[TODO after deployment: name the hosting platform, give the live URL, and add a screenshot of the app.]]

8. Limitations and Future Work

8.1 Limitations

- **Dataset shift.** The test set's class distribution differs sharply from training, so test scores describe performance on a reweighted sample, not on Kinyarwanda news in general.
- **Label noise.** Conflicting labels on identical articles and apparently inconsistent annotation (Section 6.2) place a ceiling on achievable accuracy and make some measured errors unreliable.
- **Single seed.** Each configuration was trained once. Bootstrap intervals capture test-set sampling and McNemar's test captures paired disagreement, but neither captures training variance. This matters in practice: across four complete runs of the same code, E2b's test macro-F1 varied between 0.479 and 0.509, because fastText training with several threads is not deterministic, while the TF-IDF models reproduced exactly.
- **Unequal treatment of class imbalance.** The Transformers were trained without class weights, unlike the other models, so the RQ3-versus-RQ1 comparison is not fully controlled.
- **Truncation.** 74.5% of articles exceed the Transformers' 512-token limit.
- **Scraping artefacts and encoding errors** (Section 6.2) leak into the features.
- **No out-of-domain detection.** The model always outputs a topic; the confidence warning mitigates but does not solve this.
- **One corpus and period.** Performance on newer articles or other sources has not been tested.

8.2 Future work

1. Train the Transformers with class-weighted loss, more epochs and a small learning-rate search, to make the RQ1/RQ3 comparison fair.
2. Run 3–5 seeds per configuration and report the mean and standard deviation; make fastText deterministic by training with a single thread.
3. Handle long documents with sliding windows or hierarchical encoding, so the Transformers see whole articles.

4. Evaluate a morphology-aware Kinyarwanda model such as KinyaBERT (Nzeyimana & Niyongabo Rubungo, 2022), which addresses the tokenizer limitation directly.
5. Clean the scraping artefacts and encoding errors, re-run, and measure how much of the score they carried.
6. Audit the labels of the confident mistakes with a Kinyarwanda speaker and consider multi-label annotation for overlapping topics.
7. Add explicit language identification and calibrated probabilities to the app.
8. Exploit KIRNEWS for cross-lingual transfer to Kirundi, the purpose for which the paired corpora were built (Niyongabo et al., 2020).

9. Conclusion

This project built, evaluated and prepared for deployment a 14-class topic classifier for Kinyarwanda news. Auditing the data first changed the project: more than half of the training file was duplicated, and the official test set follows a different class distribution, so the training data was deduplicated before splitting and macro-F1 was chosen as the primary metric.

The answers to the research questions are clear. Character n-grams beat word n-grams by 7.6 points with half the parameters (RQ1); fastText initialisation improved a BiLSTM by 6.7 points (RQ2); and African-centric adaptation improved a Transformer by 6.3 points (RQ3). All three improvements come from better handling of Kinyarwanda's morphology or from exposure to the language, and all three help the rarest classes most. The best model, TF-IDF character n-grams with logistic regression, reached 0.587 test macro-F1 and 72.6% accuracy, while being the cheapest model to run. Its remaining errors (RQ4) come mainly from overlapping topics, classes with only a few dozen training examples, and inconsistencies in the source labels.

The main lesson is that on a small corpus in a morphologically rich language, the choice of representation and careful data handling mattered more than model size. The Transformers' weaker results are not evidence that they cannot do better, but a reminder that controlled comparisons need every model to receive the same treatment of class imbalance and enough training.

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Software used without a dedicated citation above: pandas, NumPy and Matplotlib (versions pinned in the repository's requirements.txt).